PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

B.TECH. (CSE)

V SEMESTER

UE20303 –SOFTWARE ENGINEERING

PROJECT REPORT

ON

RESTAURANT MANAGEMENT SYSTEM

SUBMITTED BY

TEAM-13

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GitHub link: https://github.com/Nithish-9/Restaurant-Management-System.git

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PROPOSAL OF THE PROJECT

Dev tools used:

- JAVASCRIPT.
- CSS.
- HTML.
- ReactJS.
- MYSQL.
- NodeJS.
- ExpressJS.

This project helps people who have a problem of waiting in long lines in restaurants and are worried that the food items they want will get over. So through this omnifood website customer can scan the QR code and login through that and place order by viewing the menu and proceed to the payment gateway to pay. After ordering order-id and bill will be sent to the customer's mail.

Functional Features of the project are:

I. User will be able to login through the website , if they are an existing user.

II. If they are a new user then they should create an account to login into the website. Then a mail will be sent confirm their registration

III. User login information will be stored inside the database

IV. After logging into the website, user will be provided with list of food menu.

V. User can browse through all the food items and he can order the food. There is also a filter page for veg and non-veg items and in future we would like to add some more categories of food items.

VI. User will be billed with the list of food items that has been ordered.

VII. Whenever User orders the food ,it will be stored inside the database along with name of user ,total bill amount ,name of each food item ,quantity of each food item.

VIII. Then the user can proceed to the paypal payment gateway and pay the bill. After that the invoice will be sent to the customer's mail

IX. This project helps the customer to order their dishes before coming to the restaurant so that they do not have to wait for long time.

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Software Requirements Specification

VERSION 1.2

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Name	Date	Reason For Changes	Version
Nithish S	2/11/2022	Changed the database from Mongo dB to MySQL	V1.1
Pratham Hegde	6/11/2022	Added Filter option in the menu page	V1.2
Satish S	8/11/2022	Added Cart option in the menu page	V1.3

Introduction

Purpose

This document presents a detailed explanation of the objectives, features, user interface and application of Restaurant Management System in real life. It will also describe how the system will perform and under which it must operate. Both the stakeholders and the developers of the system can benefit from this document.

Intended Audience

This document is intended for different types of people such as:

- Developer
- System designer
- Tester
- Restaurant owner

This document has a sequential overview of the whole project so if a reader reads the document from top to bottom, he will get a clear idea about the project.

Product Scope

- This system will help to manage and run the restaurant business systematically. In this management system we will provide a website that can be used by the customers to order food.
- Customers can see the list of food items in the restaurant for that day. This helps them to choose the food items in advance while waiting in line
- They can order their dishes and also make payment.

References

- HTML5 - https://www.w3schools.com
- CSS3 - https://www.w3schools.com

Overall Description

Product Perspective

The Restaurant Management System helps the customer to order food at the restaurant more effectively and efficiently by placing order in advance The whole management system is designed for a general Computerized Digital Restaurant

LOGIN PAGE:

Allows user to login to the website so that they can access the features

DESCRIPTION:

Allows user to see what features and how different is the website by explaining the features.

FRONT END USER INTERFACE:

- HTML-Format the text document on the web.
- CSS: Styling sheet.
- ReactJS.

BACKEND:

- NodeJS.
- ExpressJS.

DATABASE:

- MySQL

Product Functions

- Ordering food through the app
- List of food items to search and filtering
- Sign up page to register, if already registered then directly login.
- Adding items to the cart.
- Payment-gateway.

User Classes and Characteristics

● One main type of users are people who work in mnc's or own business. They don't have much time to wait in long lines for their order in the restaurant. So they like to order in advance so that they don't have to worry about food getting over

● Another type of users are those who want to see the menu before coming to the restaurant so as to decide whether to come or not.

● There might also be users who have language problems and find it difficult to speak to the waiter. So they order in online even if they are at the restaurant.

Operating Environment

The product will be operating in a windows environment. The website shall operate in all famous browsers, for a model we are talking Microsoft Internet Explorer, Google Chrome and Mozilla Firefox. Also, it will be compatible with IE 6.0. Most of the features will be compatible with Mozilla Firefox and Opera 7.0 or higher versions. The only requirement to use this online product would be an internet connection.

The hardware configuration includes Hard Disk: of 40GB, Monitor: 15-inch Color monitor, Keyboard: 122 keys input devices required are keyboard, mouse and output devices are monitor etc.

LANGUAGE: JavaScript

It is needed to use Smart phone to place food order in this system. Which will running on Android Operating System. Whole system should be covered by Wi-Fi connection.

Assumptions and Dependencies

The assumptions are: -

● The coding should be error-free.

● The system should be user-friendly so that it is easy to use for the users.

● The system should have more capacity and provide fast access to the database.

● The website is running 24/7.

● Users may access from any computer that has internet browsing capabilities and an internet connection

● The user must have their correct usernames and passwords to enter into their online accounts and do actions.

The dependencies are: -

● The specific hardware and software due to which the product will be run.

● Based on listing requirements and specifications the project will be developed and

run.

● The end users (admin) should have a proper understanding of the product.

● The information of all users must be stored in a database that is accessible by the

website.

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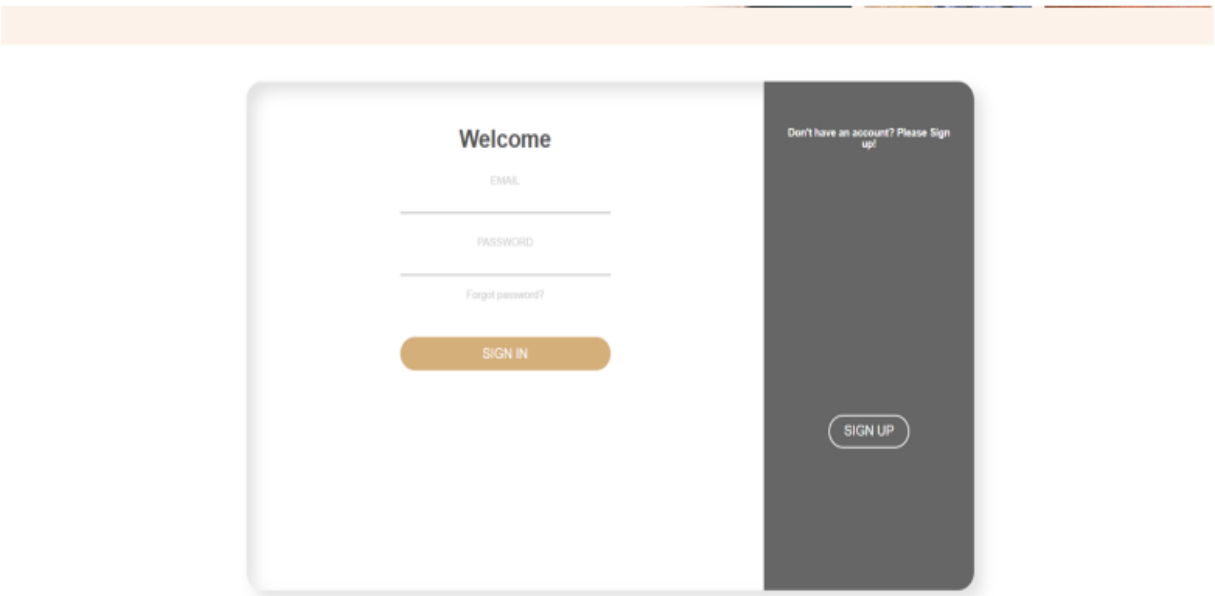
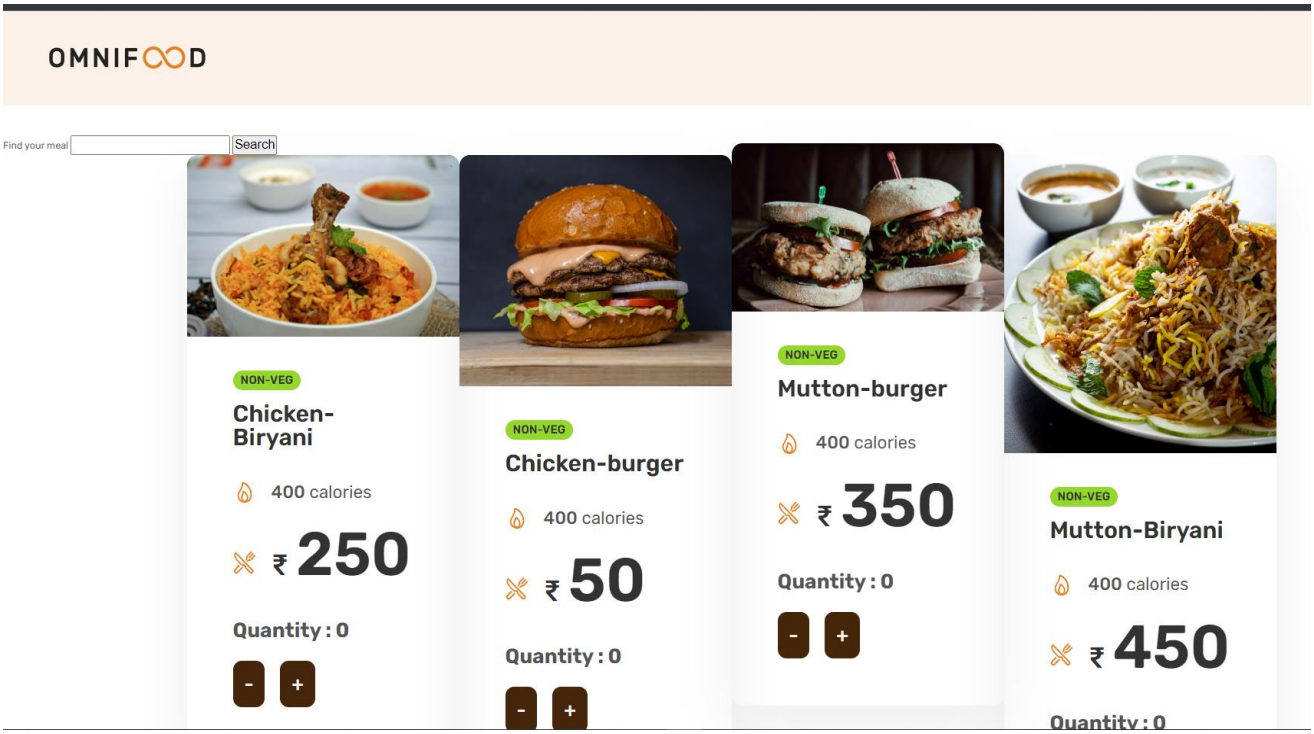
External Interface Requirements

User Interfaces

The interface will be user-friendly. So that every kind of customer can place the food order easily. Customers can also give feedback through it easily with some demo comment or if they are keen to write their review by own they can do it.
Here Customers have to login inorder to browse the food menu in the website, and then they can order the food by their own desire.

SAMPLE-SCREENSHOT





Software Interfaces:

Admin Interface:

As Admin is authorized to perform CRUD operations on menu items and prices of those items and then receiving the payment of customers.

Software Interfaces:

The software required for the development of the project is:

Operating System: Windows 10/11

Environment: Visual Studio .NET 2002

Framework: Version 1.0

Language: HTML, CSS and JavaScript

Backend: NodeJS, ExpressJS.

Database: MySQL

Communications Interfaces:

Communications Interfaces:

uses HTTP protocol for communication, so our device will follow HTTP protocol.

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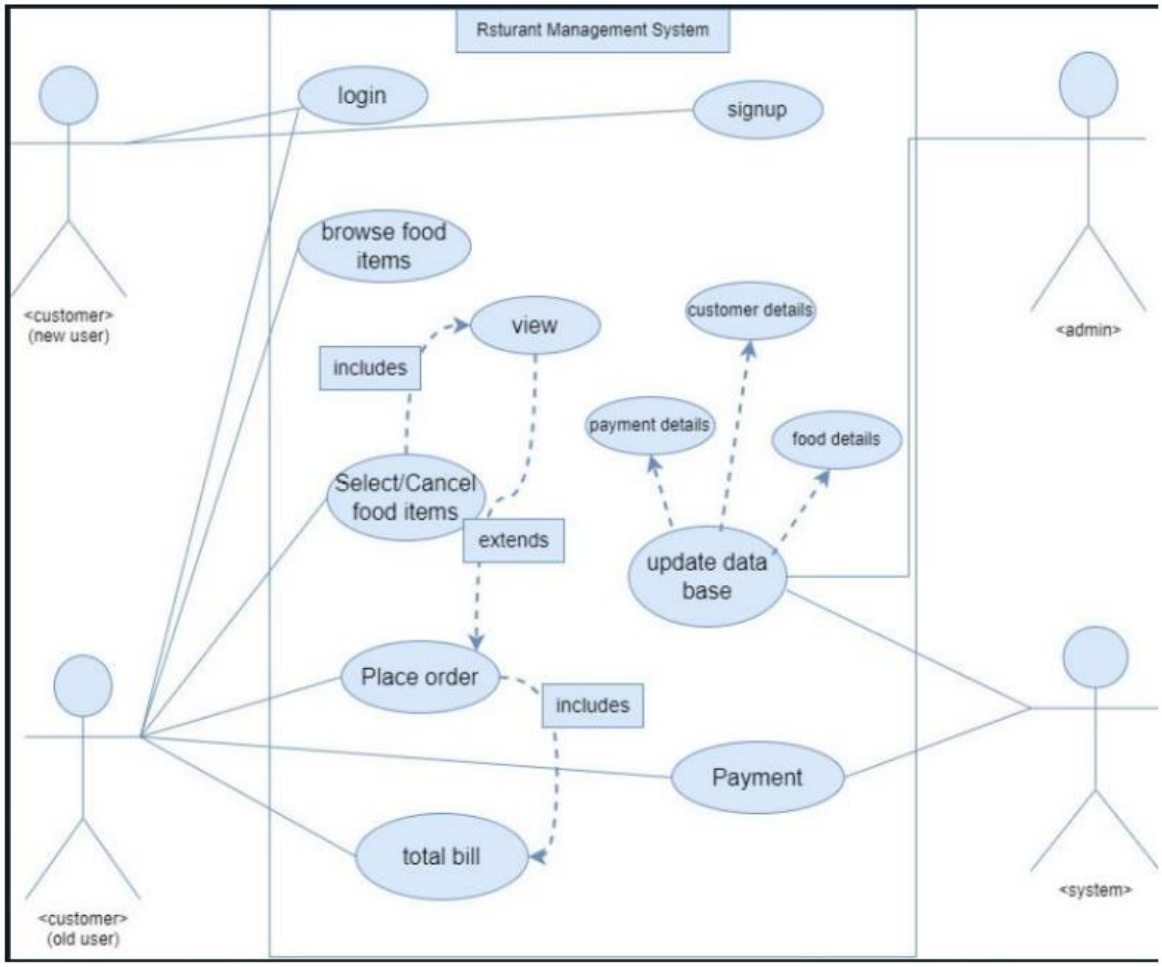
ANALYSIS MODEL:



SYSTEM FEATURES:

The Restaurant management system has the following users:
1.Customer
2.Admin
Customers are able to:
View the menu and add items to the cart

Then go to the payment page and pay
Can check for the availability of food items.
Admin can:
Check the order details and users' registration details stored in the database.

.

Other Nonfunctional Requirements

Performance Requirements

Performance based on local server.
Takes some initial load time.
Performance will depend on hardware components.
Payment system will be fully secure with POS system
Different database for each customer.

Safety Requirements

The database may get crushed at any certain time due to virus or operating system failure. Therefore, it is required to take the database backup so that the database is not lost. Proper UPS/Inverter facility should be there in case of power supply failure.

Security Requirements

Whole system is secured and only admin access the data. The system will use HTTPS, because of this protocol more secure.

Software Quality Attributes

- Only the admin can edit the database and not the users
- The quality of the database is maintained in such a way so that it can be very user friendly

Business Rules

A business rule is anything that captures and implements business policies and practices. A rule can enforce business policy, make a decision, or infer new data from existing data. This includes the rules and regulations that the system users should abide by. This includes the cost of the project. The users should avoid illegal rules and protocols. Neither the admin nor members should cross the rules and regulations.

Other Requirements

The website shall handle expected and unexpected errors in ways that prevent Loss of information and long downtime period.

Appendix A: Glossary

- User: Normally users can comment or view details of the website availability.
- Client: Intended users for the software.
- SRS: Is a document that describes what software will do how it will be expected to perform
- MySQL: is an open-source relational database management system

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PROJECT PLAN

- The life cycle we followed in this project is **Agile** as we are Focusing on Simplicity in both the product and the process.
- There is constant interaction between the stakeholders, and the requirements of the software can be changed repeatedly.
- Agile allows us to better control over the project due to constant feedback, Transparency with the clients.
- Agile works in small sprints and focuses on continuous delivery.
- This methodology works in iterations and each sprint will the better than the last one and the errors get reduced.
- Continual updates and reviews to a project's progress and requirements give you flexibility in designing the project.

For development We used languages like **HTML, CSS,JAVASCRIPT, ReactJS** to do this project.

For version control we used **GitHub.**

For designing we used **JIRA** software.

DELIVERABLES:

Creating a user-friendly interface for customers to use and order food without any difficulty.

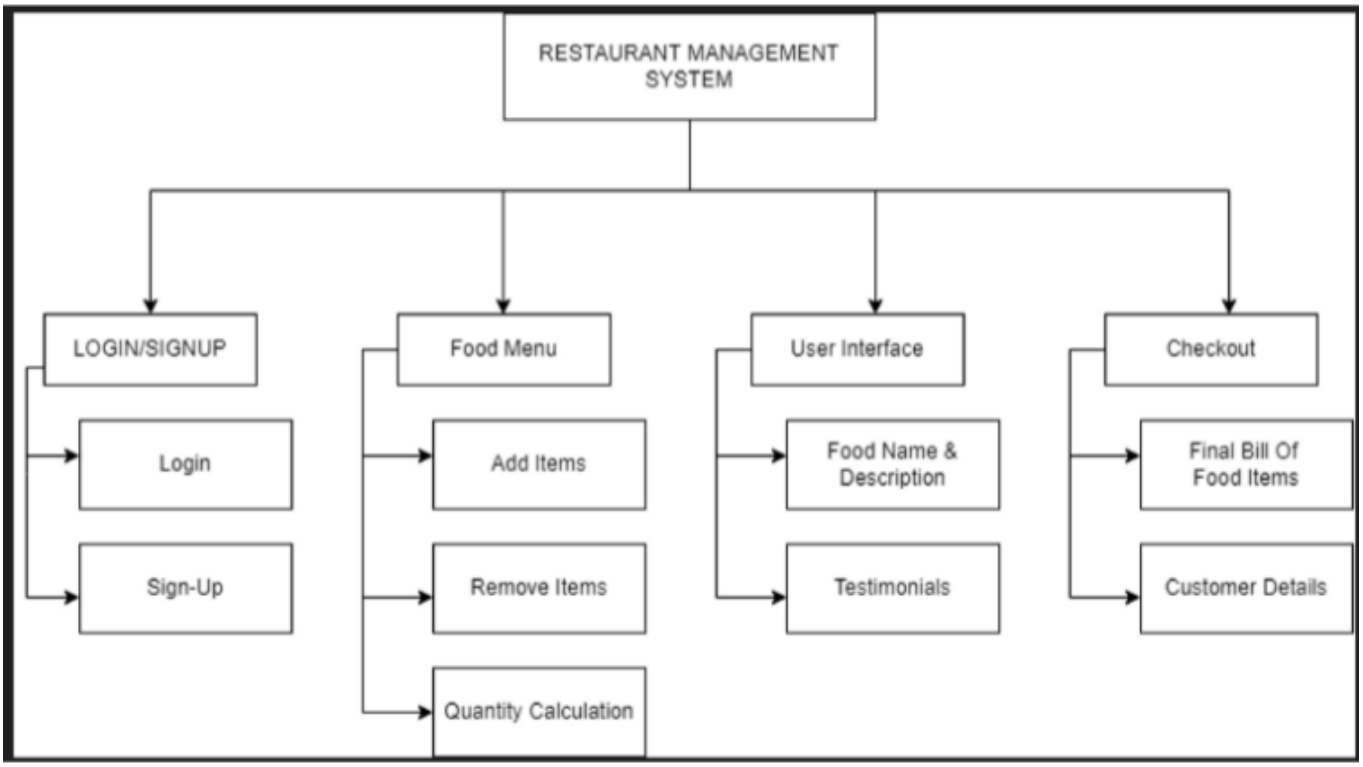
The different deliverables are the Synopsis, SRS document for the stakeholders and the client to understand functionalities, project planning document. These are all built from scratch.

Different components are:

- Login/sign up page built
- Food menu page built
- Checkout page (final bill): built
- Service page (explains the website's features): built

These are all built from scratch

WBS For the Entire Functionalities



Rough estimate of effort required to accomplish each task in terms of person months.

We can calculate this using COCOMO model.

For our project we are using organic COCOMO model as our project is:

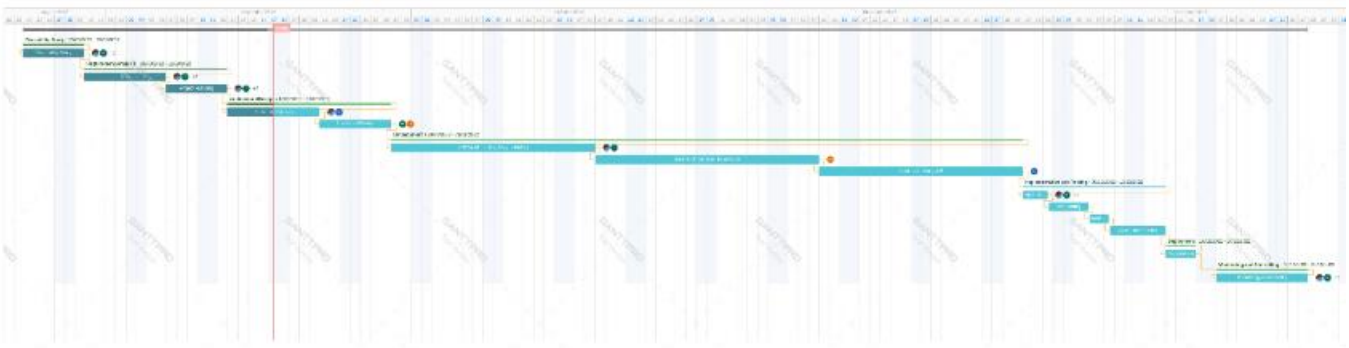
- Simple
- Team is small with prior experience
- Requirements are not rigid.

$$E = a(KLOC)^b$$

a = 2.4
b = 1.05
KLOC = 5.642

Therefore,
Effort = 14.76 Person Months.

Gant chart for scheduling the work.



	Task name	Assigned	Status
1	**Feasability Study**		
1.1	Feasability Study	P +2	Done
2	**Requirement Analy...**		
2.1	SRS	P +2	Done
2.2	Project Planning	P +2	Done
3	**Architecture/Design**		
3.1	High-Level Design	G	In progress
3.2	Low-Level Design	P N	Open
4	**Development**		
4.1	Front-End : HTM...	P	Open
4.2	Back-End : Node...	N nithishanusri0	Open
4.3	Database : Mong...	G gsneeraj2002	Open
5	**Implementation an...**		
5.1	Implementation	P +2	Open
5.2	Unit Testing	unassigned	Open
5.3	System Testing	unassigned	Open
5.4	Acceptance Testing	unassigned	Open
6	**Deployment**		
6.1	Deployment	unassigned	Open
7	**Monitoring and Co...**		

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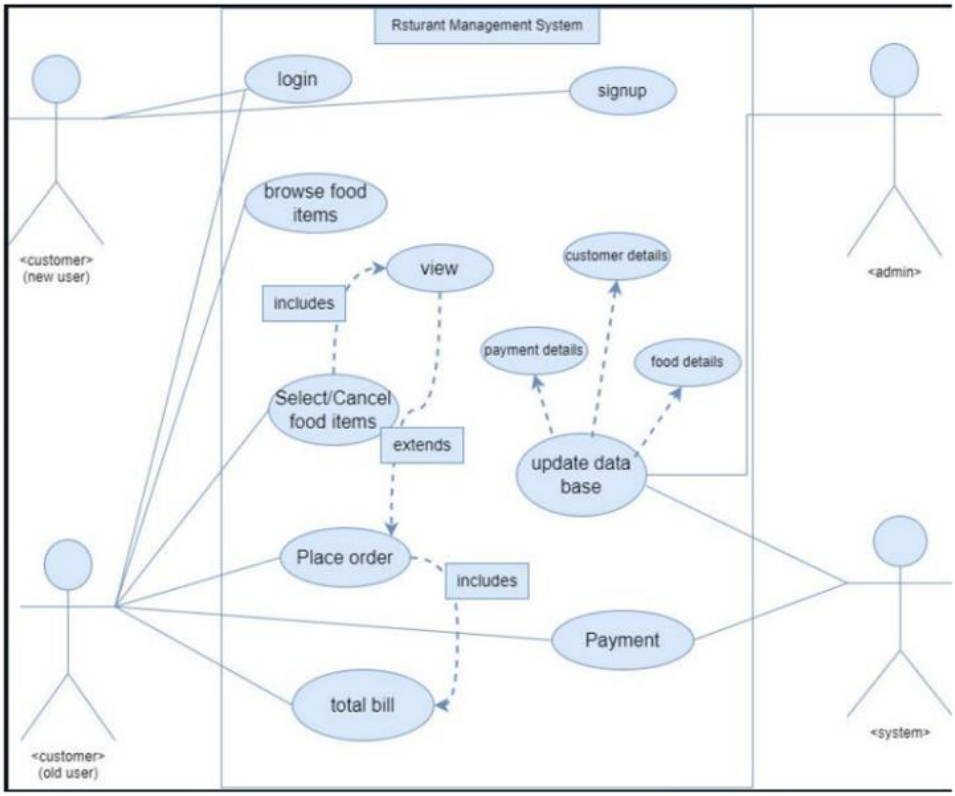
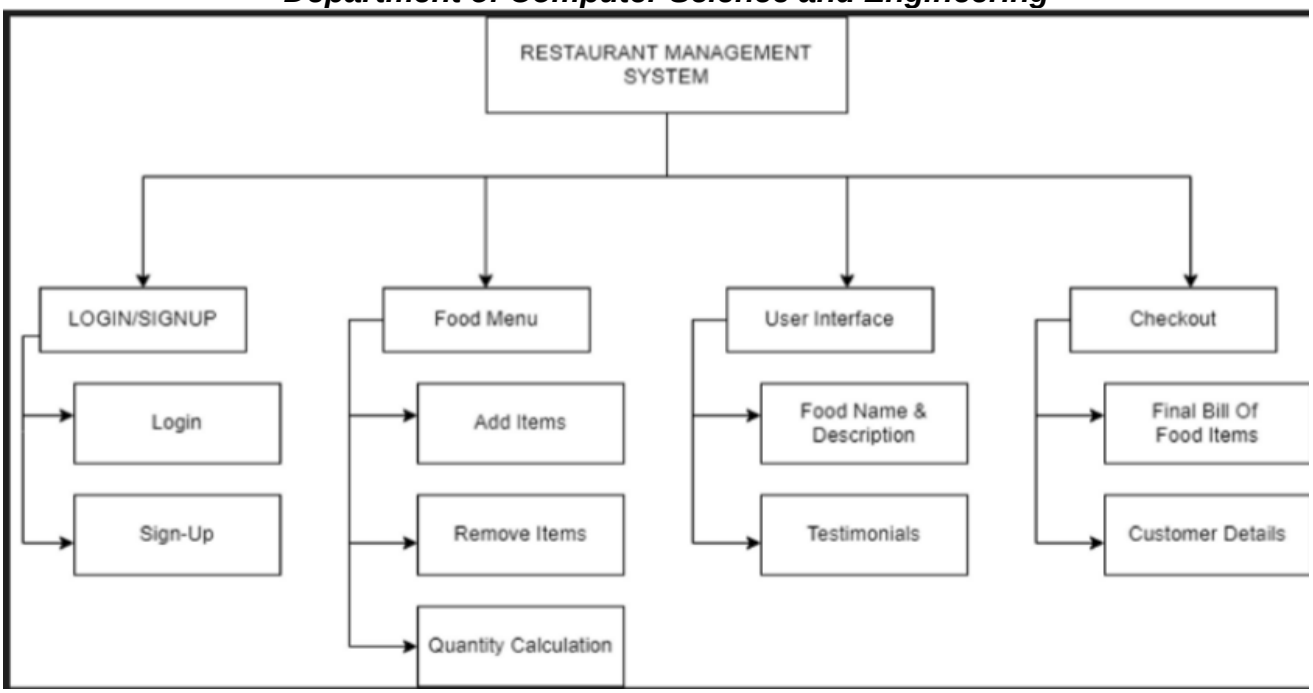
DESIGN DIAGRAM

Use-case diagram



Working structure:


RESTAURANT MANAGEMENT SYSTEM
LOGIN/SIGNUP
Login
Sign-Up
Food Menu
Add Items
Remove Items
Quantity Calculation
User Interface
Food Name & Description
Testimonials
Checkout
Final Bill Of Food Items
Customer Details

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TEST PLAN DOCUMENT

Test Case ID :

Each test case should be represented by a unique ID. To indicate test types, follow some convention like “UT_01” indicating &” Unit Testing - Test Case#1.”

- **Name of the module:**

Specify the name of the main module or sub module being tested

- **Test Case Description:**

Specify the summary or test purpose in brief

- **Pre- Conditions:** Any requirement that needs to be done before execution of this test Case.

- **Test Steps:**

Mention all the steps in detail and specify the order in which it is to be executed.

- **Test Data:**

Input for the test case to be executed. Specify different data sets with precise values to be used as input. (Create test case for both valid and invalid inputs)

- **Expected Results:**

Mention the expected results including error or precise messages that should be displayed on screen

- **Actual Results:**
  After execution of test case fill this column with the result obtained

- **Test Result (Pass/Fail):**

Mark this field as “fail” if the actual result is not same as expected result else mark as “pass”.

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Test Case ID	Name of Module	Test case description	Preconditions ns	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-01	User registration on module	To test the Functionality of homee page.	Access to Chrome Browser	Navigate to https://localhost:9000/login/signup	--	Successful display of home page with welcome message to website	Display of home page with welcome message to website	Pass
UT-02	Registration email module	To test the functionality of email module, when customer registers with valid email, the confirmation email will be sent to the customer's mail Id.	1.Access to Chrome Browser 2.Having internet access. 3. Valid mail Id	1.Navigate to https://localhost:9000/login/signup 2.Customer should check his inbox for successful registration message.	--	Successful Message of registration is sent to customers mail id.	Message of registration is sent to customers mail id.	Pass
UT-03	Food Menu Page i) Filter module	To test the functionality of displaying the various categories of food specified in our restaurant. (Ex. Veg food, Non-Veg food).	Having internet access.	1. Navigate to http://localhost:9000/rrecipes/search. 2.Choose the desired filter option in the food menu page for display of food items.		Successful display of the user chosen category food items in the filter option of food menu page.	Display of the user chosen category food items in the filter option of food menu page.	Pass
UT-04	Food Menu Page i) Cart module	To test the functionality of whether the user chosen food items added to cart along with quantity and total price of food items	Having internet access.	1. Navigate to http://localhost:9000/recipes 2. Click the increment (+) and decrement (-) button on food item in the food menu page for to get that item add on to the cart.		Successful display of food items added by the user onto the cart.	Display of food items added by the user onto the cart.	Pass
UT-05	Billing Page module	To test the functionality of all the chosen food items is added to the user's bill that he	Having internet access	1. Navigate to http://localhost:9000/recipes 2. Click the increment (+) and decrement (-) button on food item in the food menu page for to get that item add		Successful display of all the chosen food items present in the customer's	Display of all the chosen food items present in the customer's	Pass

		would continue to pay for the bill.		on to the cart. 3. Click on proceed button, which directs to billing page		bill.	bill.	
UT-06	Checkout email module	To test the functionality, whether the customer 's bill is sent the mail or not.	1.Access to Chrome Browse 2.Having internet access. 3. Valid mail Id	1.Customer should check his inbox for the total bill and ordered food items message for confirmation		Successful Message of the total bill and and ordered food items sent to customers mail id.	Message of the total bill and and ordered food items sent to customers mail id.	Pass
UT-07	Payment Module	To test the functionality of payment module.	1.Access to Chrome Browse 2.Having internet access. 3.Having a credit card or having account in any bank. 4. Having sufficient balance in the bank account	1. Navigate to http://localhost:9000/payment 2.Customer should check his details. 3.Click on Pay button and it redirects to the payment portal		Successful Display of payment portal after customer confirms with his details.	Display of payment portal after customer confirms with his details.	Pass
UT-08	Payment email module	To test the functionality of Payment email module Where the invoice of customer is sent to his mail id.	1.Access to Chrome Browse 2.Having internet access. 3. Valid mail Id	1.Customer should check his inbox for the invoice message		Successful Message of the invoice sent to customers mail id.	Message of the invoice sent to customers mail id	Pass

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Unit Testing

A unit is the smallest block of code that functions individually. The first level of testing is Unit testing and this problem statement is geared towards the same.

```html
<div class="cont">
    <div class="form sign-in">
        <h2>Welcome</h2>
        <label>
            <span>Email</span>
            <input type="email" />
        </label>
        <label>
            <span>Password</span>
            <input type="password" />
        </label>
        <p class="forgot-pass">Forgot password?</p>
        <button type="button" class="submit">Sign In</button>

    </div>
    <div class="sub-cont">
        <div class="img">
            <div class="img__text m--up">

                <h3>Don't have an account? Please Sign up!<h3>
            </div>
            <div class="img__text m--in">

                <h3>If you already has an account, just sign in.<h3>
            </div>
            <div class="img__btn">
                <span class="m--up">Sign Up</span>
                <span class="m--in">Sign In</span>
```

```
</div>
<div class="sub-cont">
    <div class="img">
        <div class="img__text m--up">

            <h3>Don't have an account? Please Sign up!<h3>
        </div>
        <div class="img__text m--in">

            <h3>If you already has an account, just sign in.<h3>
        </div>
        <div class="img__btn">
            <span class="m--up">Sign Up</span>
            <span class="m--in">Sign In</span>
        </div>
    </div>
    <div class="form sign-up">
        <h2>Create your Account</h2>
        <label>
            <span>Name</span>
            <input type="text" />
        </label>
        <label>
            <span>Email</span>
            <input type="email" />
        </label>
        <label>
            <span>Password</span>
            <input type="password" />
        </label>
```

- Develop test cases for both valid and invalid data

Test Case ID	Test Cases	Valid/Invalid	Expected	Actual
1	User with a valid username and valid password	Valid	login	Logged in
2	User with a valid username and an invalid password.	Invalid	login	Invalid details
3	User with a valid Password and an invalid username.	Invalid	login	Invalid details
4	Only password entered but username is not entered	invalid	login	It gives error message as Please fill details
5	Only username is entered but password is not entered	invalid	login	It gives error message as Please fill details.
6	Verify if there is a checkbox with the label "remember password" on the login page.	Valid	No checkbox named remember password	Check box is absent
7	closing the browser should not log-out an authenticated user.	Valid	User shouldn't be logged out of the page	Logging out by closing the browser.

DYNAMIC TESTING

Boundary Value Analysis

a) Boundary testing is a black-box (which is also called as functional testing) testing technique that software developers often use to check the errors at the boundaries or extreme ends of a given input domain.
b) An input domain consists of all the possible inputs available in a software program.
c) The extreme ends in boundary testing might include lower to upper that is lower - 1 and upper + 1.

- Build your boundary test cases and execute them

In our project min length for password is 2 and max length is 16.
so boundary conditions are (1,2,8,16,17).

Test case number	MIN characters	MAX characters	output
1	1	8	Invalid
2	2	8	Valid
3	8	8	Valid
4	16	8	Invalid
5	17	8	Invalid
6	8	1	Invalid
7	8	2	Invalid
8	8	8	Valid
9	8	16	Valid
10	8	17	Invalid

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MUTATION TESTING:

- Develop at least 3 mutants of the functioning code and test all 4 code bases using the test case from the first problem statement. In our project max = 16 and min = 2, iplen=input length.

Mutant NO	Mutant	Expected	Actual Output	Result
1	Iplen>=16 and iplen <= 2	Valid	Invalid	killed
2	Iplen < 16 and iplen > 2	Valid	Valid	alive
3	Iplen >16 and iplen	Valid	Invalid	killed

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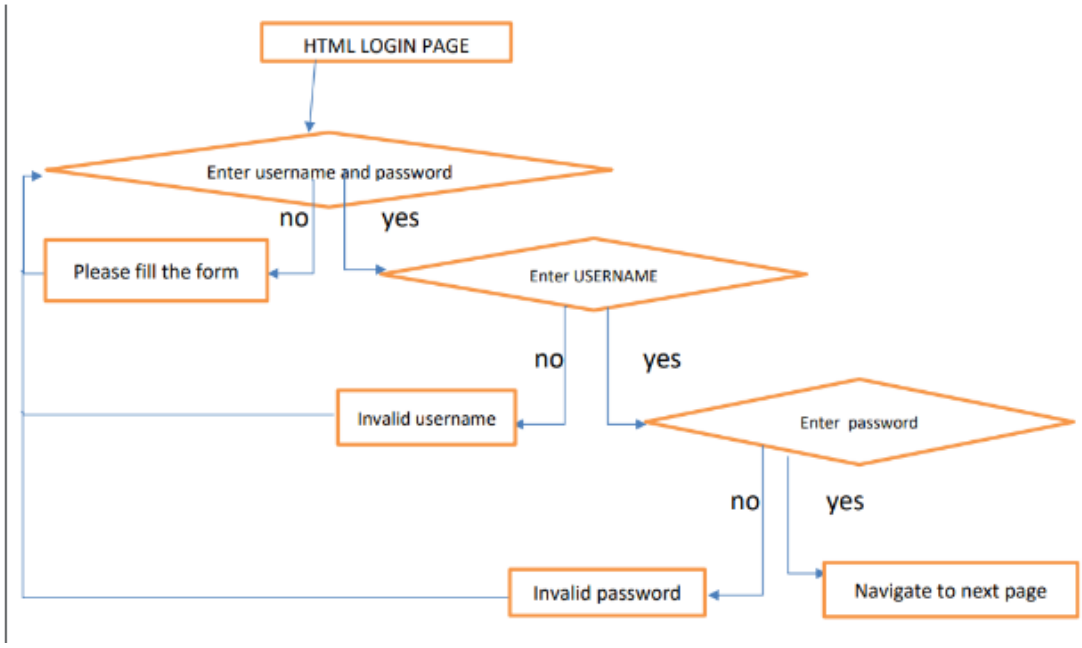
STATIC TESTING

Static testing involves validating your code without any execution. Under this problem statement, you will be expected to analyse and calculate the cyclomatic complexity of your code.

Cyclomatic Complexity Graph.



cyclomatic complexity=E-N+2P
=9-7+2(1)
=4

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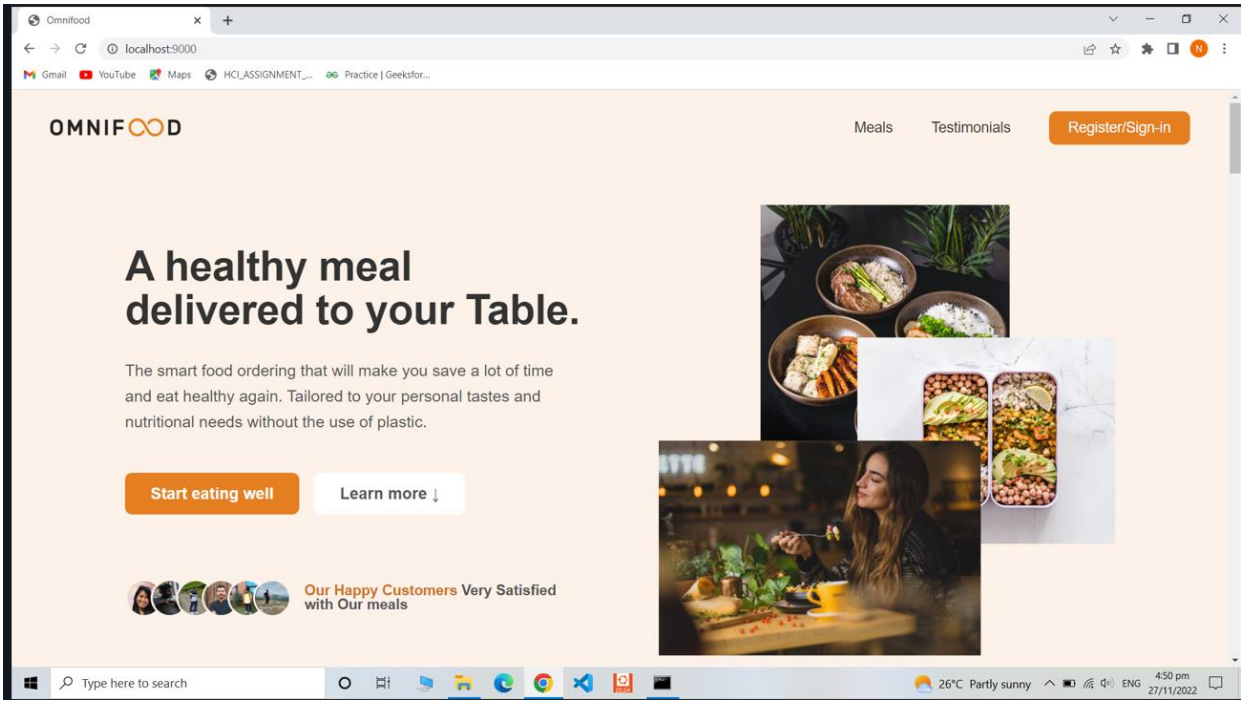
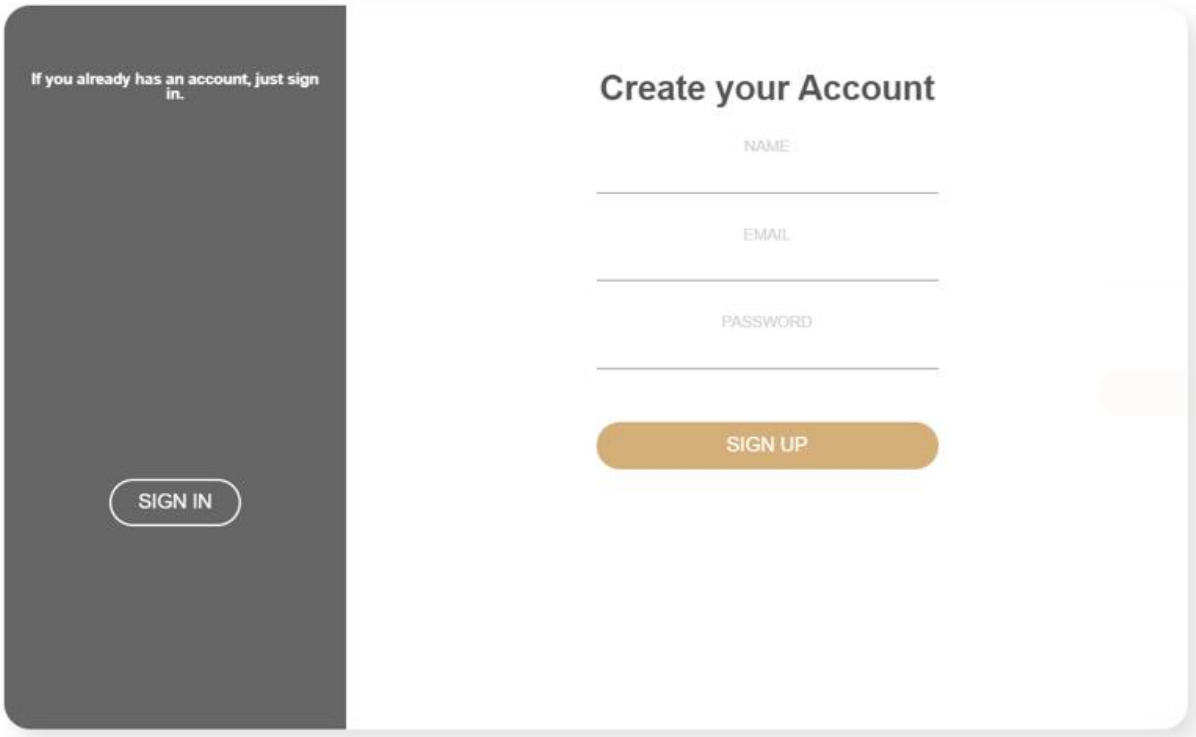
OUTPUT SCREENSHOTS:




If you already has an account, just sign in.
SIGN IN
Create your Account
NAME
EMAIL
PASSWORD
SIGN UP

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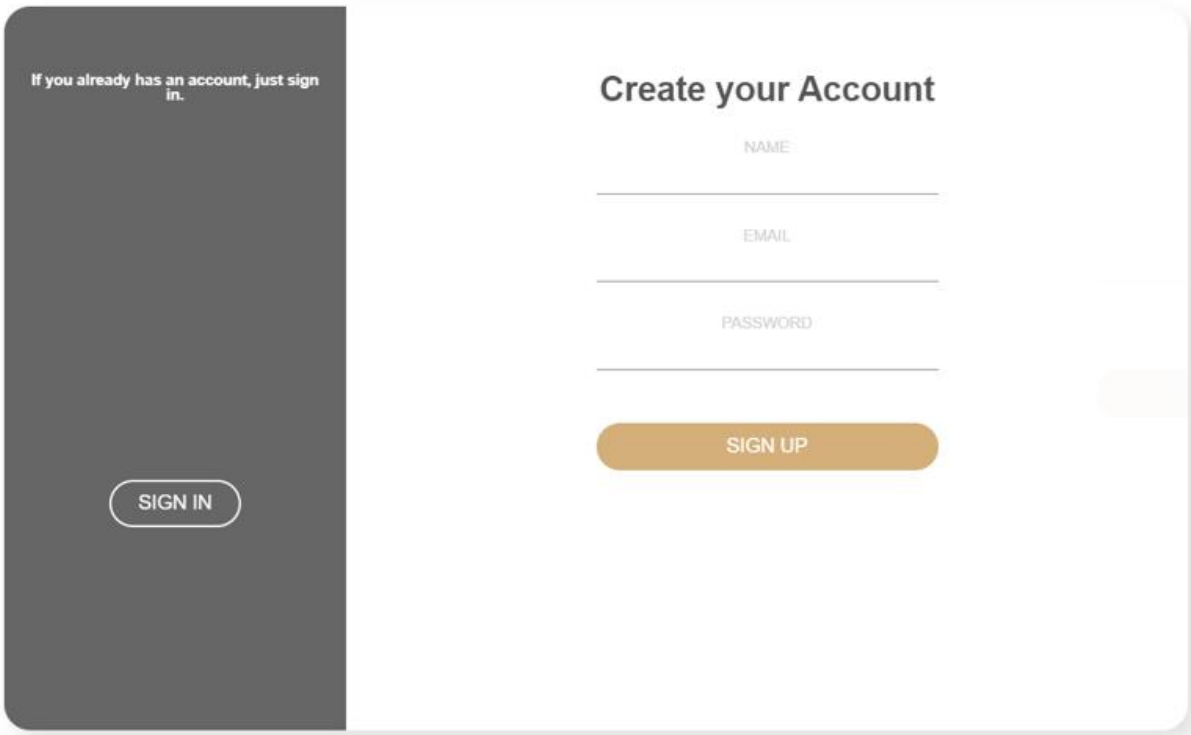

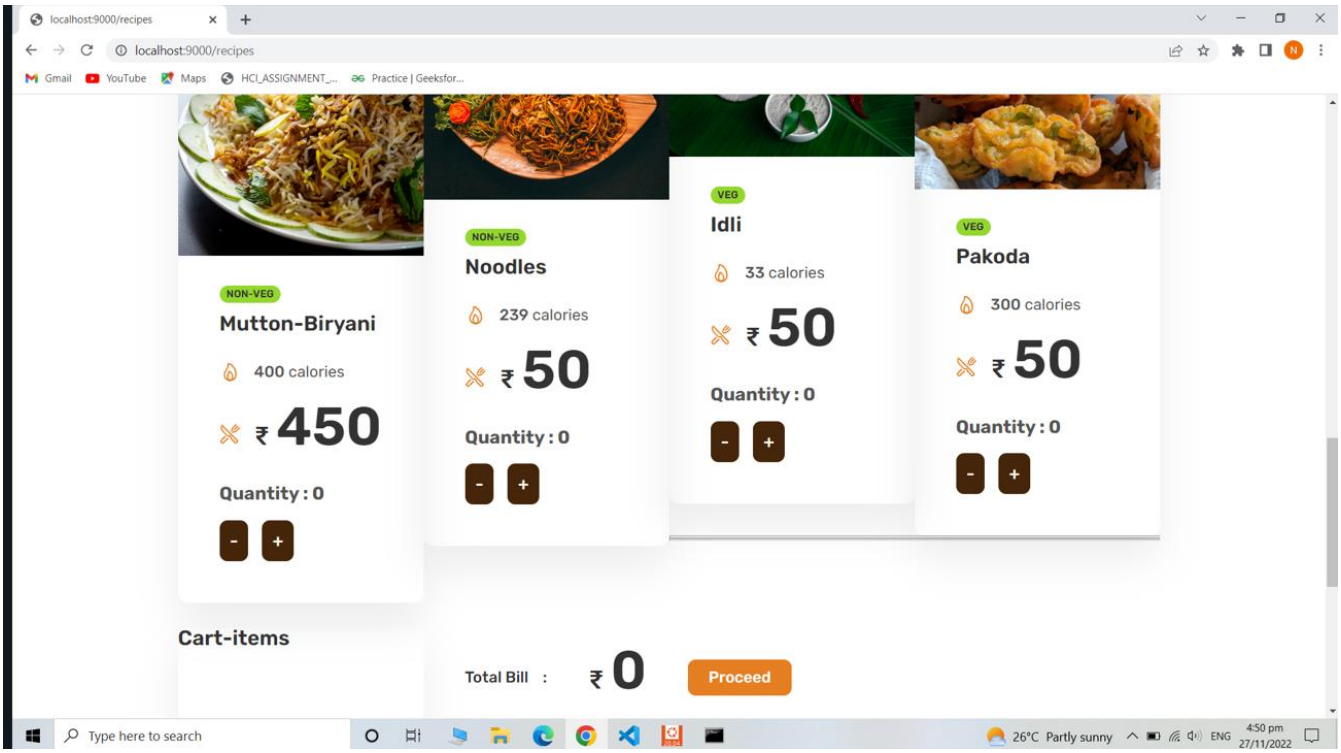

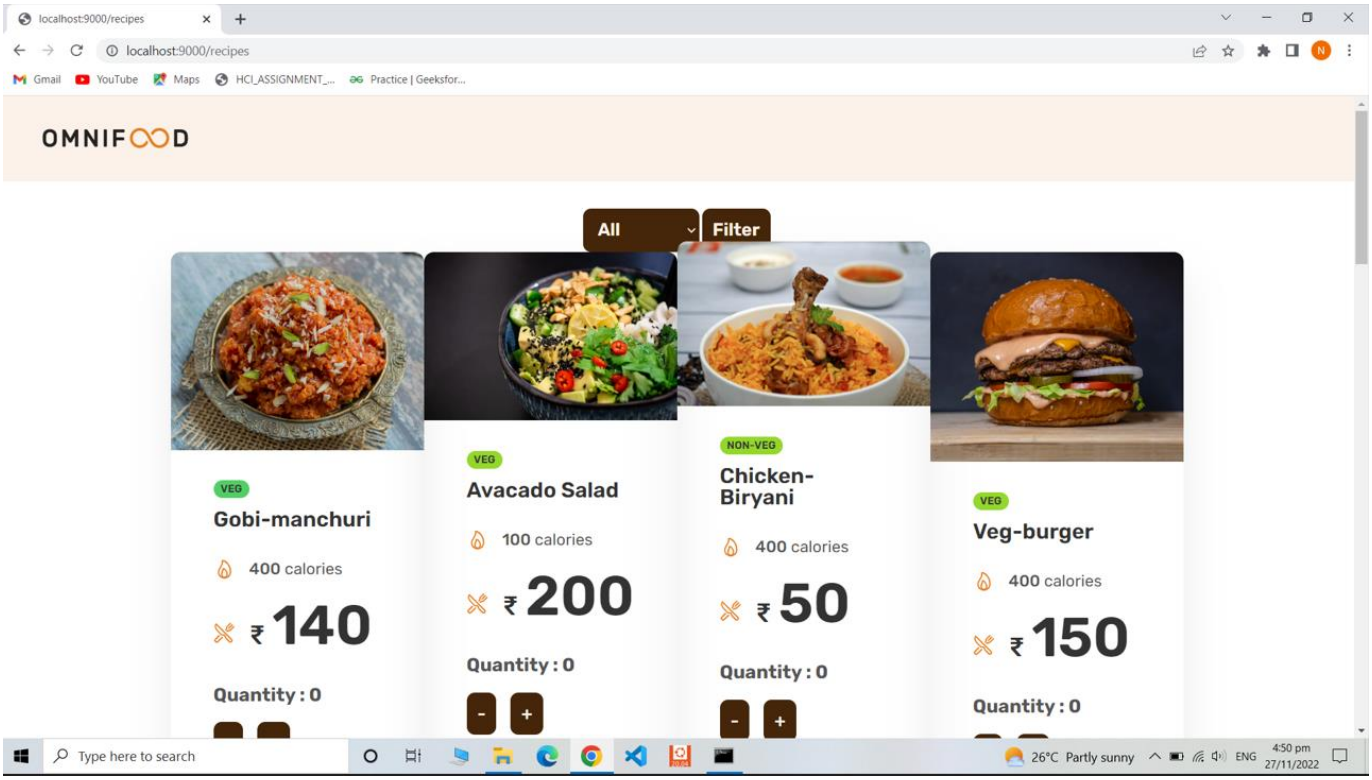

GitHub link: https://github.com/Nithish-9/Restaurant-Management-System.git

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